

OUTPUT FOR LEVEL -01

```
=====
  ATC Assignment 1 (level-1) - Identifier Recognition
=====
Enter identifiers to check (type 'exit' to quit)

Enter string: x
Result: "x" is a VALID identifier.

Enter string: _boss
Result: "_boss" is a VALID identifier.

Enter string: 1x
Result: "1x" is an INVALID identifier.

Enter string: (mahi)
Result: "(mahi)" is an INVALID identifier.

Enter string: mahi
Result: "mah" is a VALID identifier.

Enter string: ab-c
Result: "ab-c" is an INVALID identifier.

Enter string: total1
Result: "total1" is a VALID identifier.
```

OUTPUT FOR LEVEL-02

```
=====
  ATC Assignment 1 (level-2)- MiniATC Lexical Analyzer
=====
```

```
Analyzing MiniATC Program:
```

```
-----
begin
  int x;
  x = 10;
  int y;
  y = x + 5;
  print(x);
  print(y);
end
```

Line	Type	Value
1	KEYWORD	begin
2	KEYWORD	int
2	IDENTIFIER	x
2	DELIMITER	;
3	IDENTIFIER	x
3	OPERATOR	=
3	NUMBER	10
3	DELIMITER	;
4	KEYWORD	int
4	IDENTIFIER	y
4	DELIMITER	;
5	IDENTIFIER	y
5	OPERATOR	=
5	IDENTIFIER	x
5	OPERATOR	+
5	NUMBER	5
5	DELIMITER	;
6	KEYWORD	print
6	DELIMITER	(
6	IDENTIFIER	x
6	DELIMITER	)
6	DELIMITER	;
7	KEYWORD	print
7	DELIMITER	(
7	IDENTIFIER	y
7	DELIMITER	)

```
--- Token Summary ---
```

```
Keywords: 6
Identifiers: 7
Numbers: 2
Operators: 3
Delimiters: 10
```

```
--- Interactive Mode ---
```

```
Enter a MiniATC program (end with a line containing only '###'):
```

```
begin
int x;
x = 10;
print(x);
end
###
```

```
Token Table for your input:
```

Line	Type	Value
1	KEYWORD	begin
2	KEYWORD	int
2	IDENTIFIER	x
2	DELIMITER	;
3	IDENTIFIER	x
3	OPERATOR	=
3	NUMBER	10
3	DELIMITER	;
4	KEYWORD	print
4	DELIMITER	(
4	IDENTIFIER	x
4	DELIMITER	)
4	DELIMITER	;
5	KEYWORD	end

```
--- Token Summary ---
```

```
Keywords: 4
Identifiers: 3
Numbers: 1
Operators: 1
Delimiters: 5
```

OUTPUT FOR LEVEL - 03

```
=====
ATC Assignment 1 (Level-3) - MiniATC Lexer + Parser
=====
```

Source Program:

```
-----
begin
  int x;
  x = 10;
  int y;
  y = x + 5;
  print(x);
  print(y);
end
-----
```

```
[Parsing: Program]
  Matched 'begin'
[Parsing: StmtList]
[Parsing: Stmt]
  [DeclStmt]
    Matched 'int'
    Matched identifier: x
    Matched ';'
  [Parsing: Stmt]
    [AssignStmt]
      Matched identifier: x
      Matched '='
  [Parsing: Expr]
  [Parsing: Term]
  [Parsing: Factor]
    Matched number: 10
    Matched ';'
  [Parsing: Stmt]
    [DeclStmt]
      Matched 'int'
      Matched identifier: y
      Matched ';'
  [Parsing: Stmt]
    [AssignStmt]
      Matched identifier: y
      Matched '='
```

```
    [AssignStmt]
      Matched identifier: y
      Matched '='
  [Parsing: Expr]
  [Parsing: Term]
  [Parsing: Factor]
    Matched identifier: x
  [Parsing: ExprTail with '+']
  [Parsing: Term]
  [Parsing: Factor]
    Matched number: 5
    Matched ';'
  [Parsing: Stmt]
    [PrintStmt]
      Matched 'print'
      Matched '('
      Matched identifier: x
      Matched ')'
      Matched ';'
  [Parsing: Stmt]
    [PrintStmt]
      Matched 'print'
      Matched '('
      Matched identifier: y
      Matched ')'
      Matched ';'
      Matched 'end'
```

```
===== Parse Result =====
SUCCESS: Program is syntactically CORRECT.
=====
```

```

Source Program:
-----
begin
  int x
  x = 10;
end
-----

[Parsing: Program]
  Matched 'begin'
[Parsing: StmtList]
[Parsing: Stmt]
  [DeclStmt]
    Matched 'int'
    Matched identifier: x
    [ERROR] Line 3: Expected ';', found 'x'
[Parsing: Stmt]
  [ERROR] Line 3: Unexpected token '='
[Parsing: Stmt]
  [ERROR] Line 3: Unexpected token '10'
[Parsing: Stmt]
  [ERROR] Line 3: Unexpected token ';'
  Matched 'end'

===== Parse Result =====
      FAILED: 4 syntax error(s) found.
=====

```

```

--- Interactive Parser Mode ---
Enter MiniATC program (end line with '###'):
begin
int y;
y = x + 5;
print(y);
end
###

```

```

Source Program:
-----
begin
int y;
y = x + 5;
print(y);
end
-----

[Parsing: Program]
  Matched 'begin'
[Parsing: StmtList]
[Parsing: Stmt]
  [DeclStmt]
    Matched 'int'
    Matched identifier: y
    Matched ';'
[Parsing: Stmt]
  [AssignStmt]
    Matched identifier: y
    Matched '='
[Parsing: Expr]
[Parsing: Term]
[Parsing: Factor]
  Matched identifier: x
[Parsing: ExprTail with '+']
[Parsing: Term]
[Parsing: Factor]
  Matched number: 5
  Matched ';'
[Parsing: Stmt]
  [PrintStmt]
    Matched 'print'

```

```
[Parsing: Program]
  Matched 'begin'
[Parsing: StmtList]
[Parsing: Stmt]
  [DeclStmt]
    Matched 'int'
    Matched identifier: y
    Matched ';'
  [Parsing: Stmt]
    [AssignStmt]
      Matched identifier: y
      Matched '='
  [Parsing: Expr]
  [Parsing: Term]
  [Parsing: Factor]
    Matched identifier: x
  [Parsing: ExprTail with '+']
  [Parsing: Term]
  [Parsing: Factor]
    Matched number: 5
    Matched ';'
  [Parsing: Stmt]
    [PrintStmt]
      Matched 'print'
      Matched '('
      Matched identifier: y
      Matched ')'
      Matched ';'
      Matched 'end'

===== Parse Result =====
      SUCCESS: Program is syntactically CORRECT.
=====
```